

Example run — STimage → HEST pipeline

Clean.py deduplicates STimage against HEST, then Assemble.py converts the safe slides.

```
(base) bradzap@Brads-MacBook-Air ~ % conda activate hest
(hest) bradzap@Brads-MacBook-Air ~ % python3 "/Users/bradzap/Developer/GitHub/Hest Assembling Pipeline/Clean.py"
/Users/bradzap/Developer/GitHub/STimage-1K4M/meta/meta_all_gene.csv
Where do you wish to store cleaned metadata? Please input directory: /Users/bradzap/Developer
Notice: metadata will be stored under newly created folder: /Users/bradzap/Developer/meta
Cleaning...
Total STimage slides : 1149
Confirmed duplicates : 123 (skipped)
Ambiguous          : 80  → ambiguous_metadata.csv
Safe to convert     : 946 → cleaned_metadata.csv

Ambiguous slides are in: /Users/bradzap/Developer/meta/ambiguous_metadata.csv
Please open it, review each slide, and delete any rows that are
confirmed duplicates of HEST samples. Save the file when done.

Merge remaining ambiguous rows into cleaned_metadata.csv? [Y/N]: N
Ambiguous rows not added. cleaned_metadata.csv unchanged.
(hest) bradzap@Brads-MacBook-Air ~ % python3 "/Users/bradzap/Developer/GitHub/Hest Assembling Pipeline/Assemble.py"
/Users/bradzap/Developer/meta/cleaned_metadata.csv
Where do you wish to store the data? Please input directory: /Users/bradzap/Developer
Notice: data will be stored under newly created folder: /Users/bradzap/Developer/data
This might take a long time. Converting...
[1/946] GSE144239_GSM4284316 (ST)
GSE144239_GSM4284316_coord.csv: 39.4kB [00:00, 23.6MB/s]
ST/gene_exp/GSE144239_GSM4284316_count.c(...): 100%|██████████| 45.9M/45.9M [00:01<00:00, 45.5MB/s]
ST/image/GSE144239_GSM4284316.png: 100%|██████████| 11.8M/11.8M [00:00<00:00, 12.5MB/s]
objc[27700]: Class GNotificationCenterDelegate is implemented in both ...libopenslide.1.dylib and ...libgio-2.0.0.dylib.
One of the duplicates must be removed or renamed.
saving to pyramidal tiff... can be slow
.../hest/HESTData.py:1657: FutureWarning: Use `squidpy.pl.spatial_scatter` instead.
    fig = sc.pl.spatial(adata, show=False, img_key="downscaled_fullres", color=[key], ...)
.../trident/segmentation_models/model_zoo/otsu.py:49: FutureWarning: Parameter `min_size` is deprecated since 0.26.0; use
`max_size` instead.
    otsu_masking = sk_morphology.remove_small_objects(otsu_masking, 60)
.../trident/segmentation_models/model_zoo/otsu.py:56: FutureWarning: Parameter `area_threshold` is deprecated since
0.26.0; use `max_size` instead.
    otsu_masking = sk_morphology.remove_small_holes(otsu_masking, 5000)
✓ done
[2/946] GSE144239_GSM4284317 (ST)
GSE144239_GSM4284317_coord.csv: 38.1kB [00:00, 89.0MB/s]
ST/gene_exp/GSE144239_GSM4284317_count.c(...): 100%|██████████| 45.1M/45.1M [00:00<00:00, 55.9MB/s]
ST/image/GSE144239_GSM4284317.png: 100%|██████████| 11.8M/11.8M [00:00<00:00, 14.1MB/s]
^CTraceback (most recent call last):
  File ".../Hest Assembling Pipeline/Assemble.py", line 159, in <module>
    convert_slide(slide_name, result["stimage_dir"], result["hest_dir"], technology)
  File ".../Hest Assembling Pipeline/Assemble.py", line 72, in convert_slide
    img_array = np.array(Image.open(os.path.join(stimage_dir, f"{slide_name}.png"))).convert("RGB"))
  File ".../PIL/Image.py", line 1070, in convert
    self.load()
  File ".../PIL/ImageFile.py", line 412, in load
    n, err_code = decoder.decode(b)
KeyboardInterrupt
(hest) bradzap@Brads-MacBook-Air ~ %
```

■ prompt ■ typed command ■ your input / warnings ■ output ■ traceback

The KeyboardInterrupt at slide 2 is a manual Ctrl-C, not an error. Re-running Assemble.py resumes: slide 1 is skipped (already has aligned_adata.h5ad) and slide 2 restarts from its cached download.